X-Road Lab Report

MOSD ↔ MOHA Identity Exchange over X-Road 7.7.0

4 October 2026 · Motlatsi Victor Mpahane · Local Docker Compose lab, instance LS-DEV

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1. Summary

On 4 October 2026 I rebuilt a local X-Road 7.7.0 test network and got the Ministry of Social Development (MOSD) Grants Portal fetching citizen data from the Ministry of Home Affairs (MOHA) Population Registry, entirely over X-Road. All three test calls (REST record lookup, REST vital status, SOAP identity check) returned HTTP 200, and the MOHA audit log recorded each one with the verified X-Road client `LS-DEV/GOV/MOSD/GRANTS-PORTAL`.

The network has one Central Server (cs), three Security Servers (ss1 for MCST management services, ss2 for MOSD, ss3 for MOHA), a test CA with OCSP and time-stamping, and the two ministry information systems. Each information system can reach only its own Security Server, so every request had to pass through ss2 → ss3.

2. Environment and identifiers

Instance `LS-DEV`, member class `GOV`, X-Road 7.7.0 in Docker Compose on Ubuntu, all ports bound to 127.0.0.1.

Component	Owner	Identifier	Admin / URL
Central Server (cs)	X-Road operator	`LS-DEV`	https://cs.lab:4000
ss1	Ministry of Communications, Science and Technology	`LS-DEV:GOV:MCST:SS1`	https://ss1.lab:1000
ss2	Ministry of Social Development	`LS-DEV:GOV:MOSD:SS2`	https://ss2.lab:2000
ss3	Ministry of Home Affairs	`LS-DEV:GOV:MOHA:SS3`	https://ss3.lab:3000
Management services	MCST	`LS-DEV:GOV:MCST:MANAGEMENT`	hosted on ss1
Grants Portal (consumer IS)	MOSD	`LS-DEV:GOV:MOSD:GRANTS-PORTAL`	http://localhost:8500
Population Registry (provider IS)	MOHA	`LS-DEV:GOV:MOHA:POPULATION-REGISTRY`	http://localhost:8001/audit
Test CA, OCSP, TSA	—	Test CA	http://ca:8888 (OCSP), http://ca:8899 (TSA)

The Population Registry has no user screens; its audit page is the only browser view, by design.

3. Architecture

Three diagrams describe the lab: where each container runs and how the networks separate them (Diagram 1), what happens on one eligibility check (Diagram 2), and how a Security Server joins the network through ss1 (Diagram 3). The full docker-compose.yml is in Appendix A.

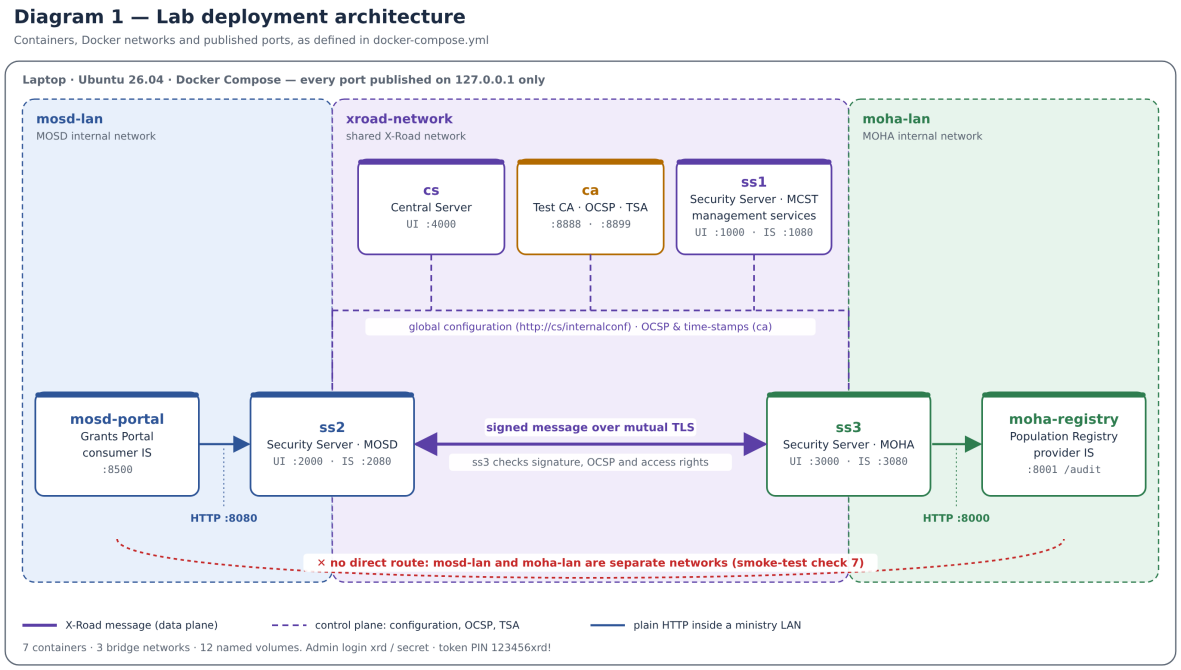


Diagram 1. *Lab deployment architecture: 7 containers on 3 Docker networks. The portal can reach the registry only through ss2 and ss3.*

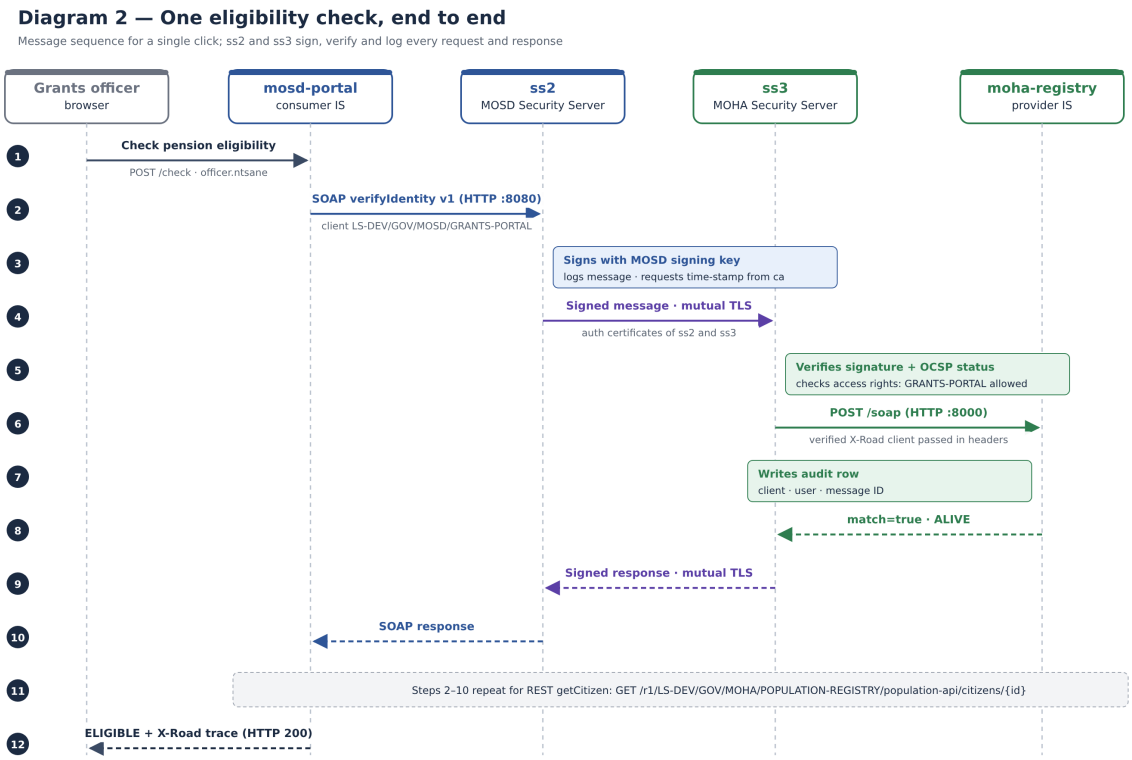


Diagram 2. *Message sequence for one eligibility check: signed and verified on both Security Servers, logged at the registry.*

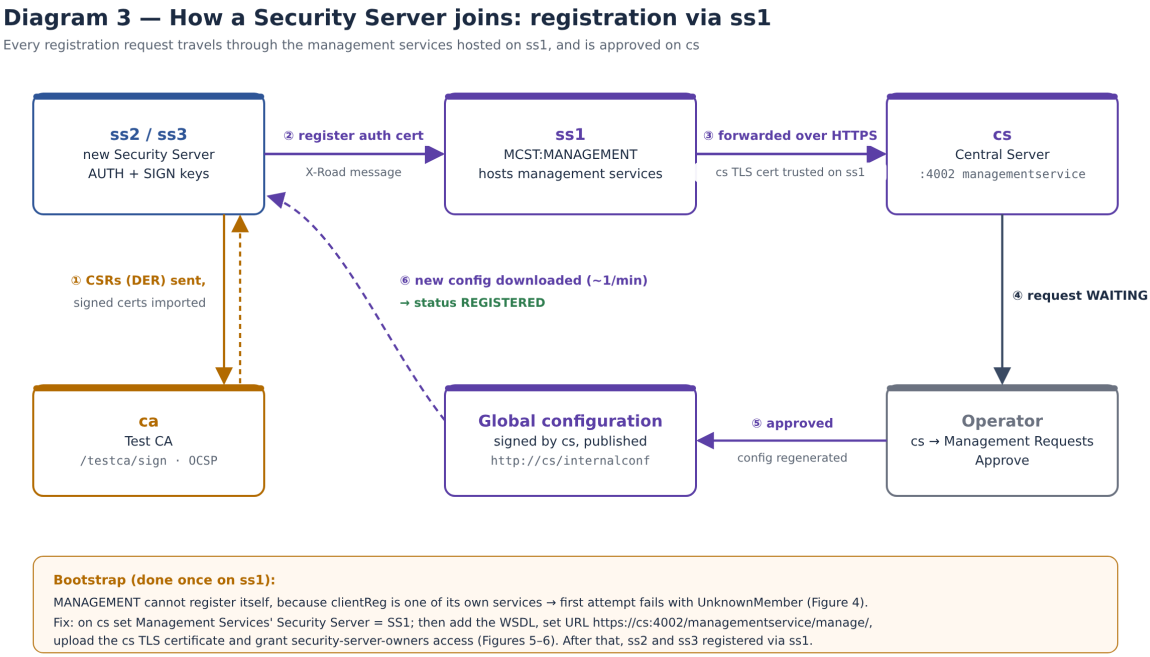


Diagram 3. *Registration flow: requests from ss2 and ss3 travel through the management services on ss1 and are approved on cs.*

4. Work done

All six phases of the lab guide were completed, in this order.

1. **Central Server.** Initialised cs with instance `LS-DEV` and address `cs`. Added member class `GOV`, the Test CA with its OCSP responder, and the Test TSA. Added members MCST, MOSD and MOHA, each with one subsystem. Set the management service provider to `MCST:MANAGEMENT`, added internal and external signing keys, and downloaded the configuration anchor.
2. **ss1 (management Security Server).** Initialised with owner `GOV:MCST`, server code `SS1`. Added the TSA, generated SIGN and AUTH keys and CSRs (DER), signed them at the Test CA, imported and activated the certificates, and registered the AUTH certificate. Registered MANAGEMENT by setting SS1 as the Management Services' Security Server on cs. Added the management WSDL, set the service URL `https://cs:4002/managementservice/manage/`, installed the cs TLS certificate, and gave `security-server-owners` access to all management services.
3. **ss2 (MOSD) and ss3 (MOHA).** Repeated initialisation, keys, CSRs, certificate import and AUTH registration with owners `GOV:MOSD` and `GOV:MOHA`; approved both on cs.
4. **IS subsystems.** Registered `POPULATION-REGISTRY` on ss3 and `GRANTS-PORTAL` on ss2; approved both on cs.
5. **MOHA services on ss3.** Added REST `population-api` from `http://moha-registry:8000/openapi.yaml` and SOAP `verifyIdentity` v1 from the WSDL. Enabled both and granted access to `LS-DEV:GOV:MOSD:GRANTS-PORTAL`.
6. **Testing.** Ran the three calls from the Grants Portal and confirmed them on the MOHA audit page.

5. Management requests on cs

All six registration requests were approved; cs now lists SS1, SS2 and SS3.

Id	Created	Type	Server	Status
6	2026-10-04 20:51	Client registration (GRANTS-PORTAL)	LS-DEV:GOV:MOSD:SS2	Approved

Id	Created	Type	Server	Status
5	2026-10-04 20:50	Client registration (POPULATION-REGISTRY)	LS-DEV:GOV:MOHA:SS3	Approved
4	2026-10-04 20:44	Auth certificate registration	LS-DEV:GOV:MOHA:SS3	Approved
3	2026-10-04 20:31	Auth certificate registration	LS-DEV:GOV:MOSD:SS2	Approved
2	2026-10-04 20:05	Client registration (MANAGEMENT)	LS-DEV:GOV:MCST:SS1	Approved
1	2026-10-04 12:20	Auth certificate registration	LS-DEV:GOV:MCST:SS1	Approved

6. Test results

Test applicant: Mpho Mokoena, national ID 010119550001, born 1955-01-01 (synthetic data); officer `officer.ntsane`.

Call	API	Result in portal	HTTP	Time	Message ID
verifyIdentity v1	SOAP	Found: yes; details match: yes; ALIVE	200	159 ms	c968d4a9-fea3-484b-8318-a0f333fa2591
getVitalStatus	REST	ALIVE (minimal disclosure)	200	187 ms	LS-DEV-23bcc6fd-07f5-4b84-9629-686b327ed6…
getCitizen	REST	Full record: F, Maseru · Ha Thetsane, LS, ALIVE	200	594 ms	LS-DEV-91a9d6c8-48b2-4ce8-a100-0f0ea2d341b0

The MOHA audit page logged all three calls (19:10–19:12) with X-Road client `LS-DEV/GOV/MOSD/GRANTS-PORTAL`. That client ID comes from the message ss2 signed and ss3 verified, so the registry can trust who is asking without its own login.

7. Problems and fixes

What I saw	Cause	Fix
ss1: `Signer.InternalError`, MCST:MANAGEMENT has no suitable certificates (Figure 1)	MANAGEMENT registered before the owner's certificates were in place	Finished owner registration, then retried
ss1: `ClientProxy.UnknownMember` on MANAGEMENT/clientReg (Figure 4)	Expected: MANAGEMENT cannot register through services not yet hosted	Set Management Services' Security Server to SS1 on cs
CSR fails with `Signer.InternalError`	Comma in Organization Name	Removed the comma
cs: configuration signing / generation fails	Signing keys or management provider missing	Added both keys; set provider MCST:MANAGEMENT
ss1: TSA Add greyed out	Test CA / TSA not yet on cs	Added on cs, waited 2 min, refreshed

8. Evidence: screenshots in order

The 17 screenshots below, taken between 19:36 and 21:13 on 4 October 2026, follow the build from ss1 management services through to the end-to-end test.

Phase 2 — ss1 and management services

Figure 1. ss1: first MANAGEMENT registration attempt fails (`Signer.InternalError`, no suitable certificates), before the owner certificates were ready.

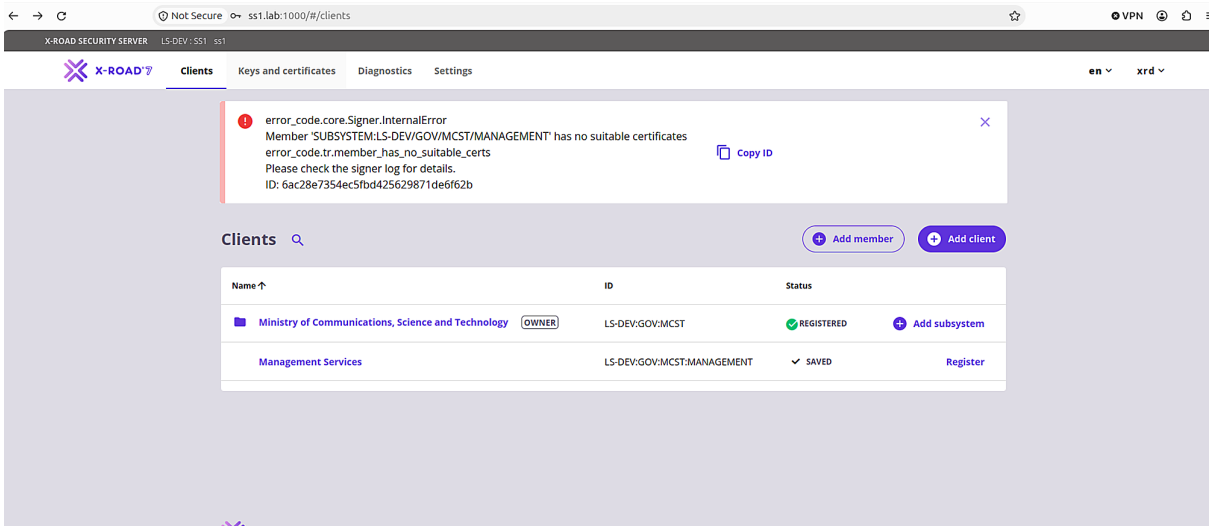


Figure 2. ss1: SIGN and AUTH certificates imported and REGISTERED.

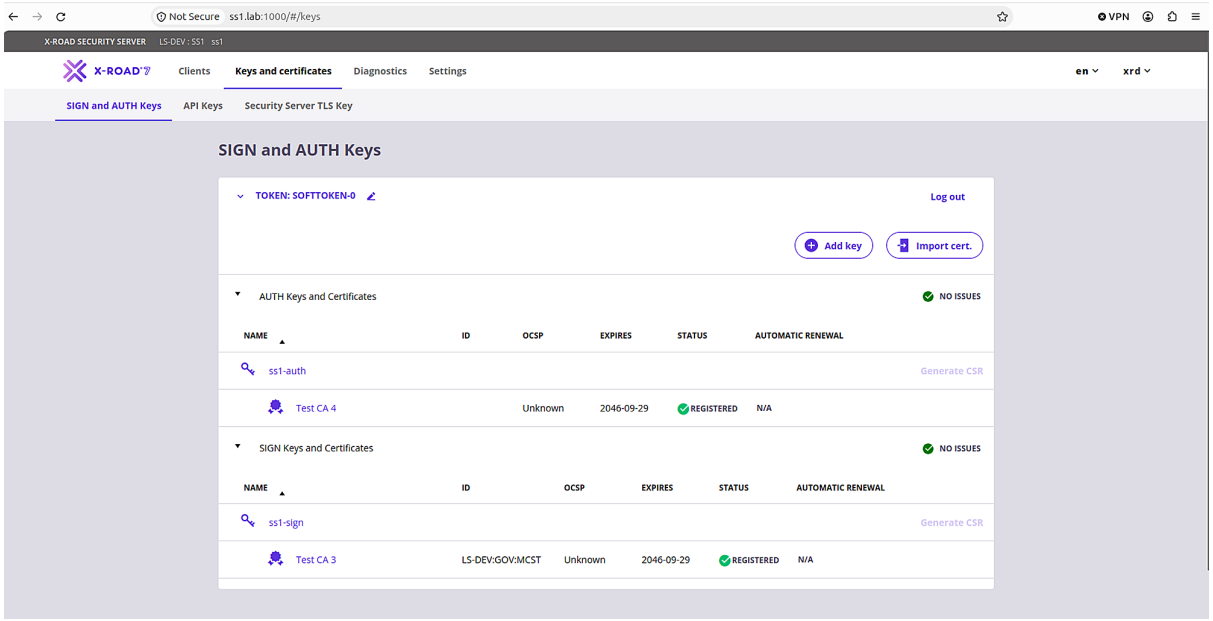


Figure 3. ss1 Diagnostics: global configuration "Everything ok"; time-stamping and OCSP waiting for their first request.

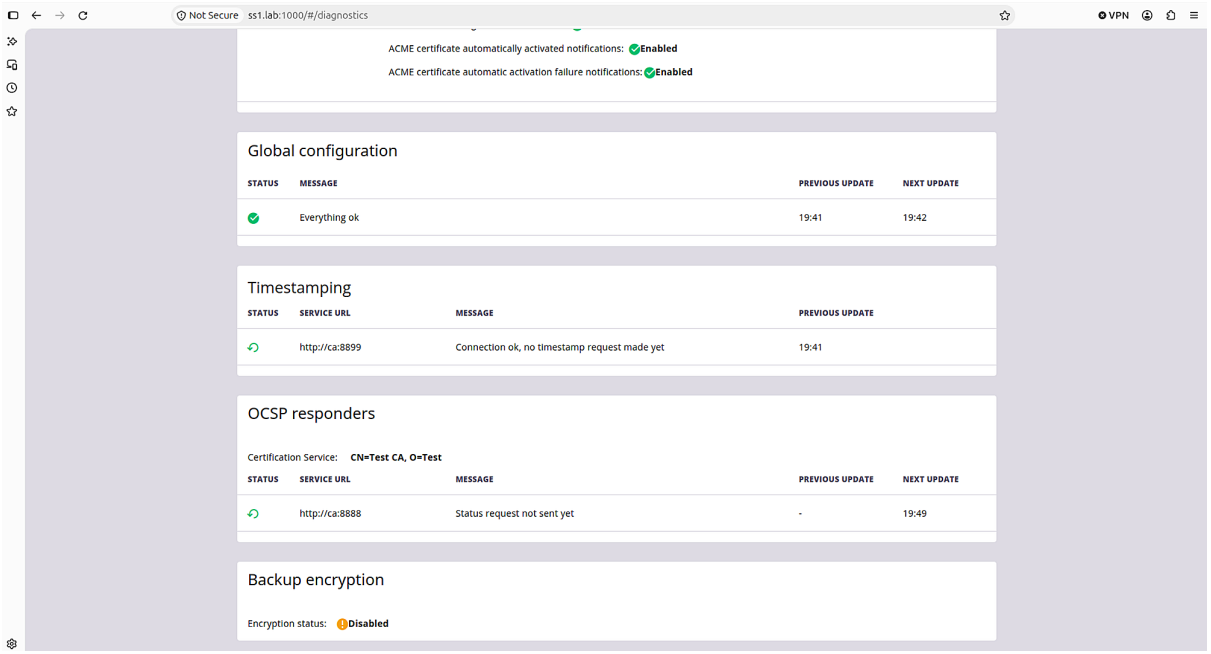


Figure 4. ss1: expected `ClientProxy.UnknownMember` on MANAGEMENT clientReg, because the management services were not yet hosted.

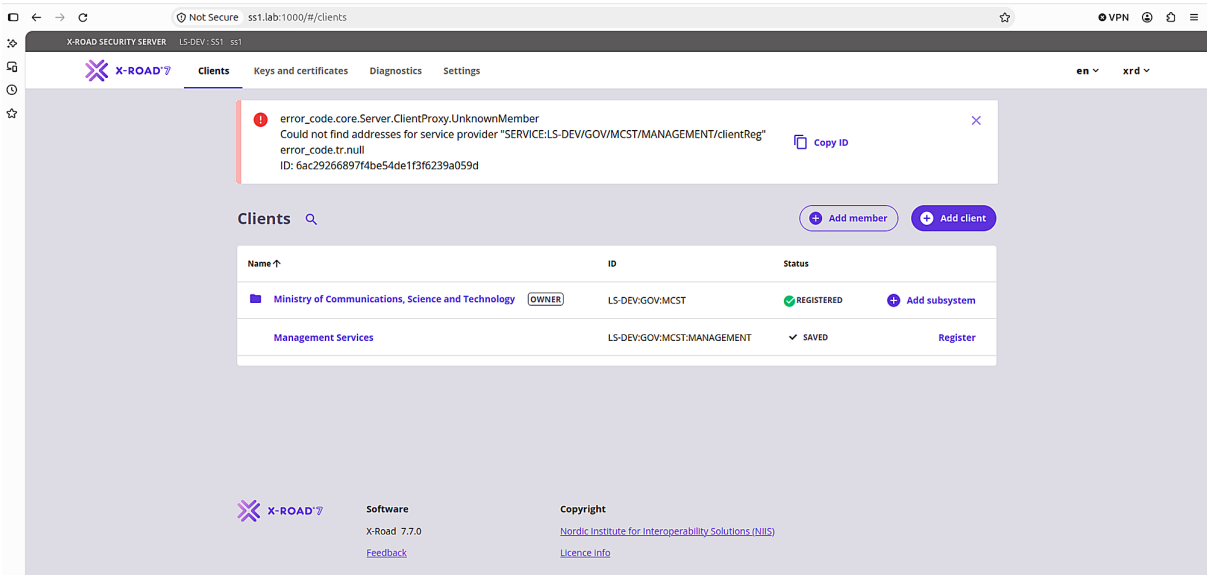


Figure 5. cs: Management Services settings. Provider is MCST:MANAGEMENT; the Management Services' Security Server was then set to SS1.

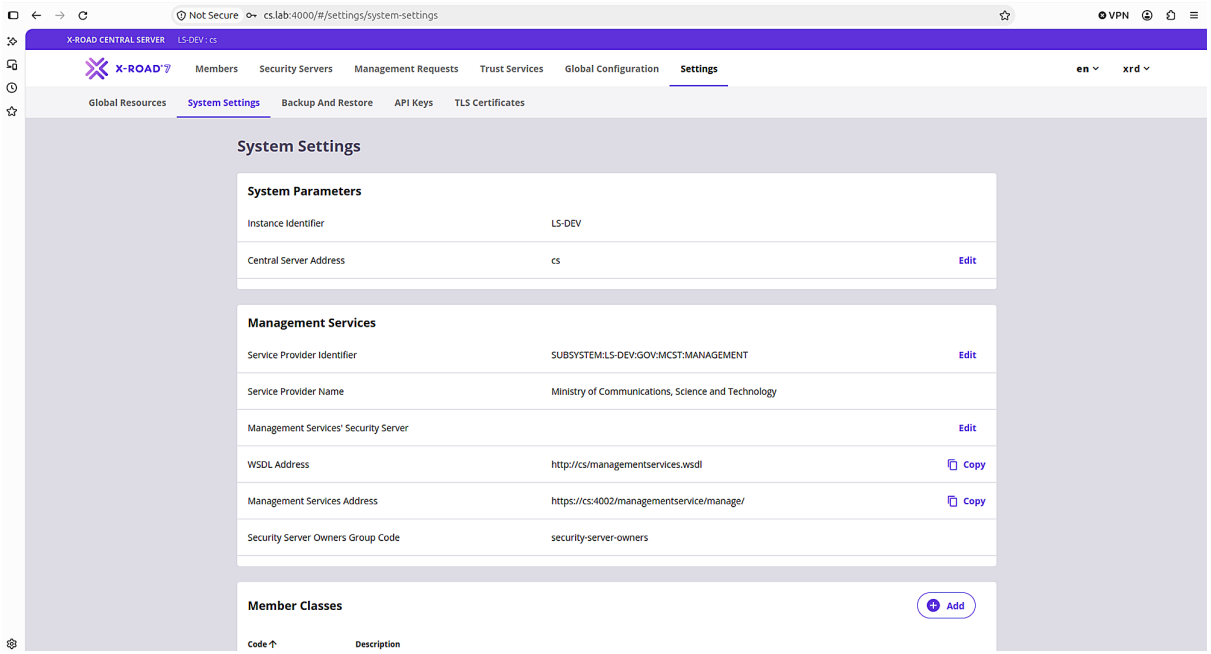
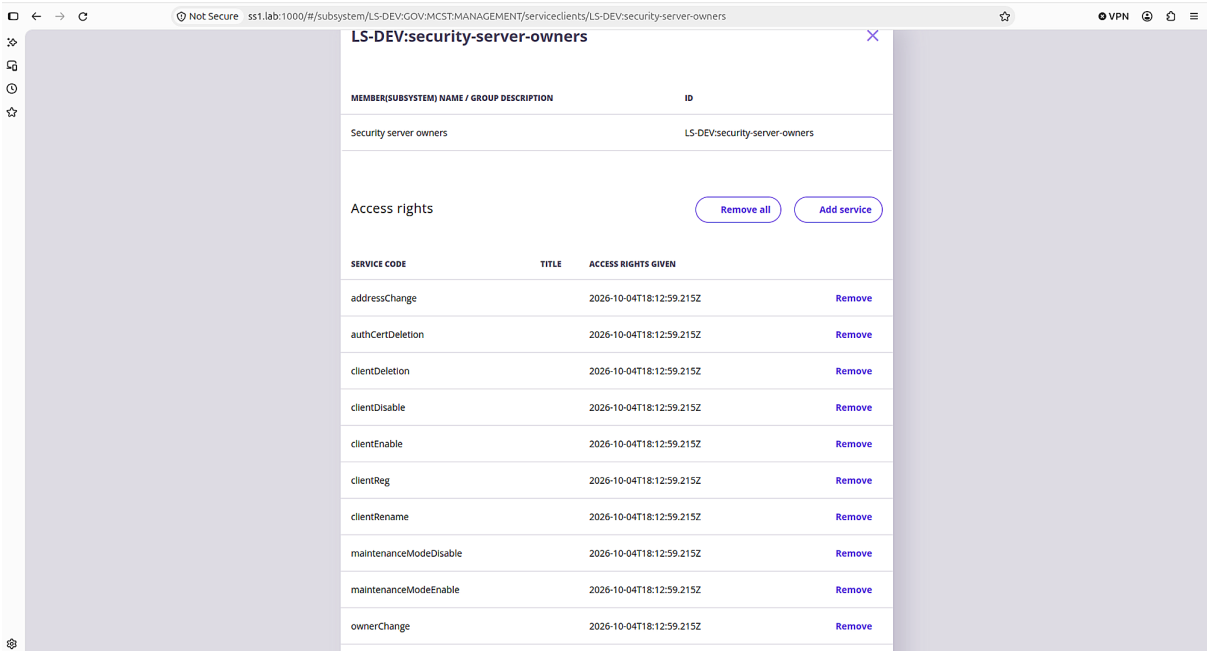
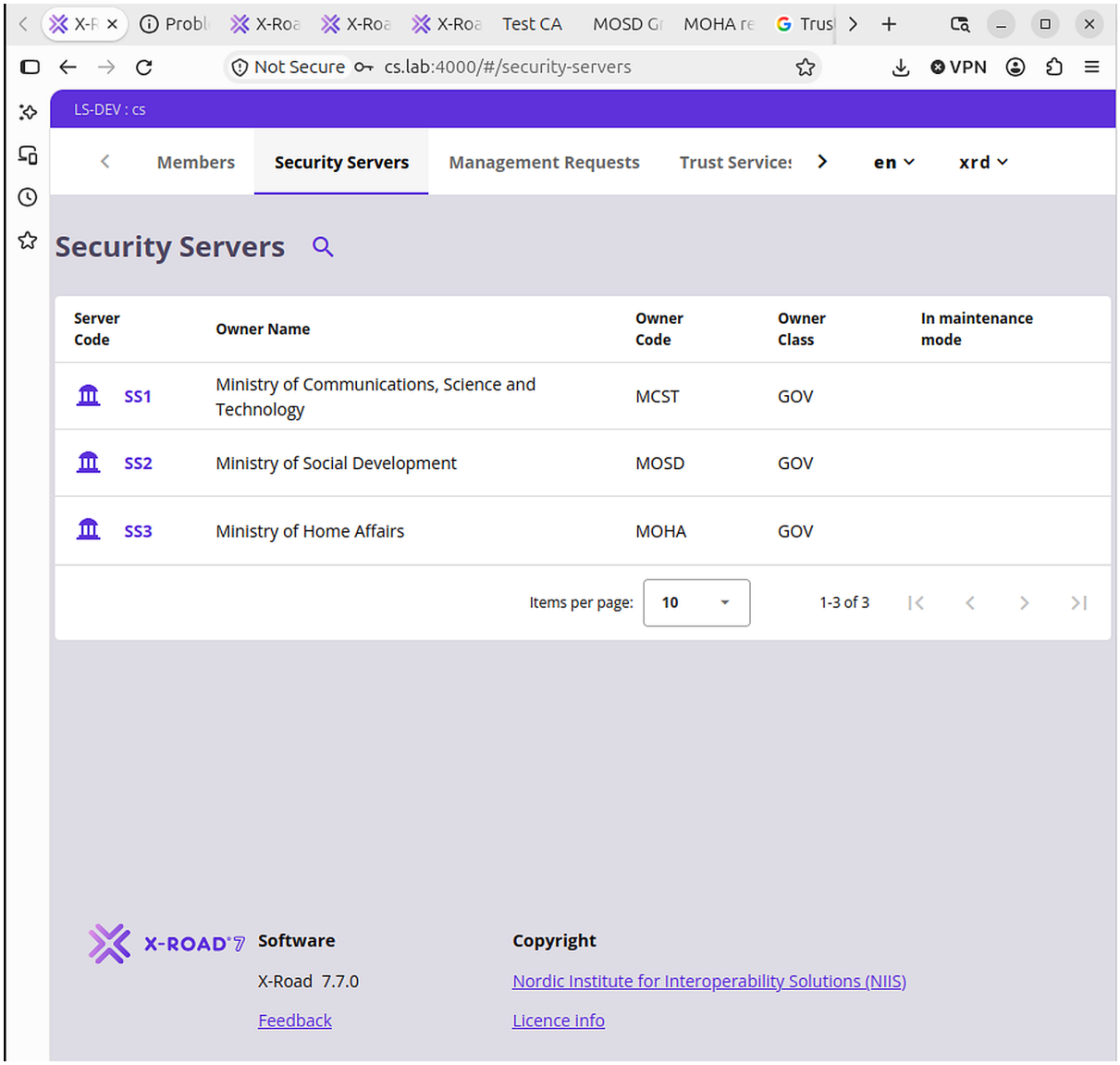


Figure 6. ss1: `security-server-owners` given access to the management services.



Phase 3 — ss2 and ss3

Figure 7. cs: SS1, SS2 and SS3 all registered.



Phase 4 — IS subsystems

Figure 8. ss3: adding the POPULATION-REGISTRY subsystem with Register ticked.

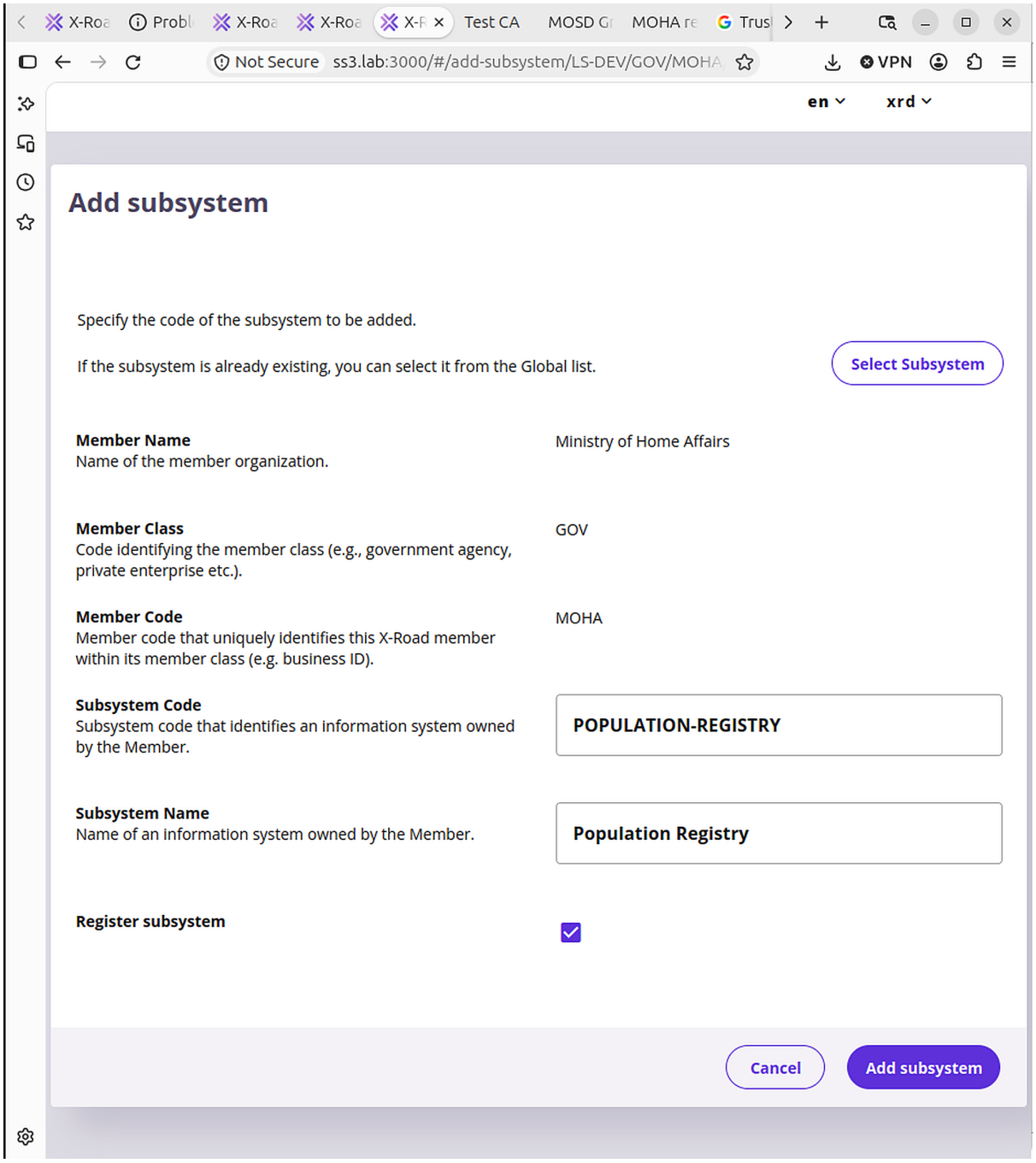


Figure 9. cs: all six management requests APPROVED.

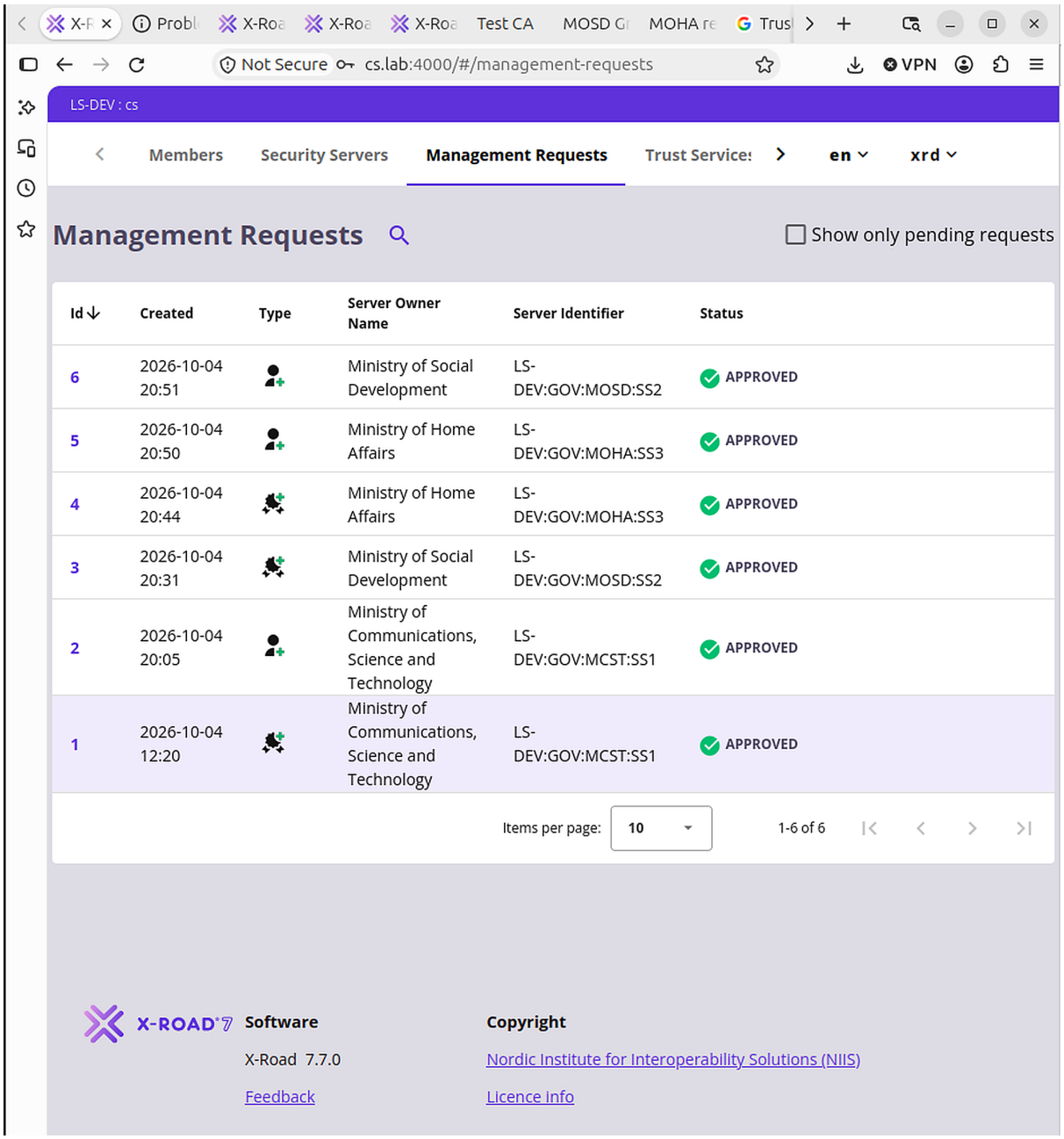


Figure 10. ss3: Population Registry "Registration in progress".

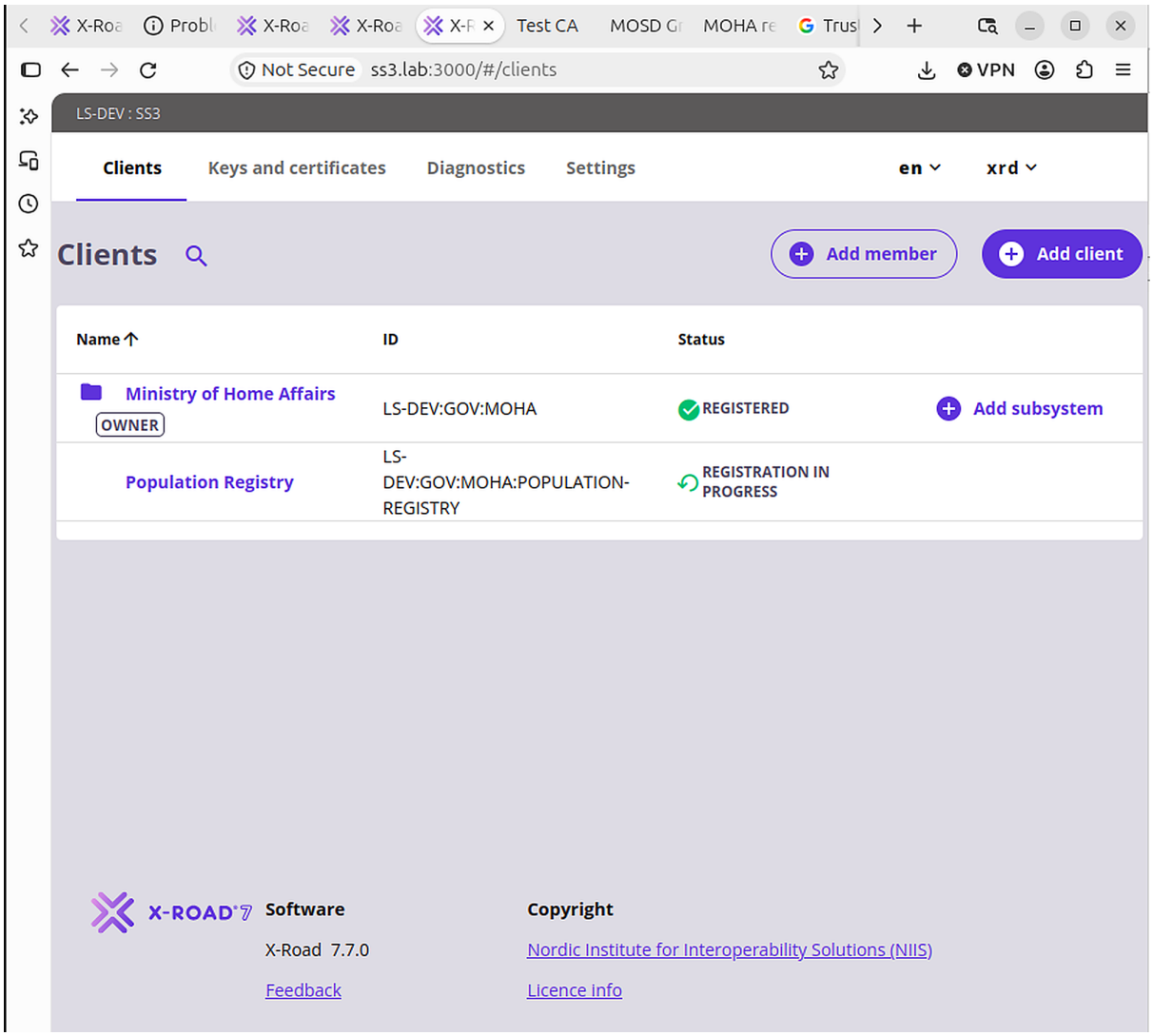
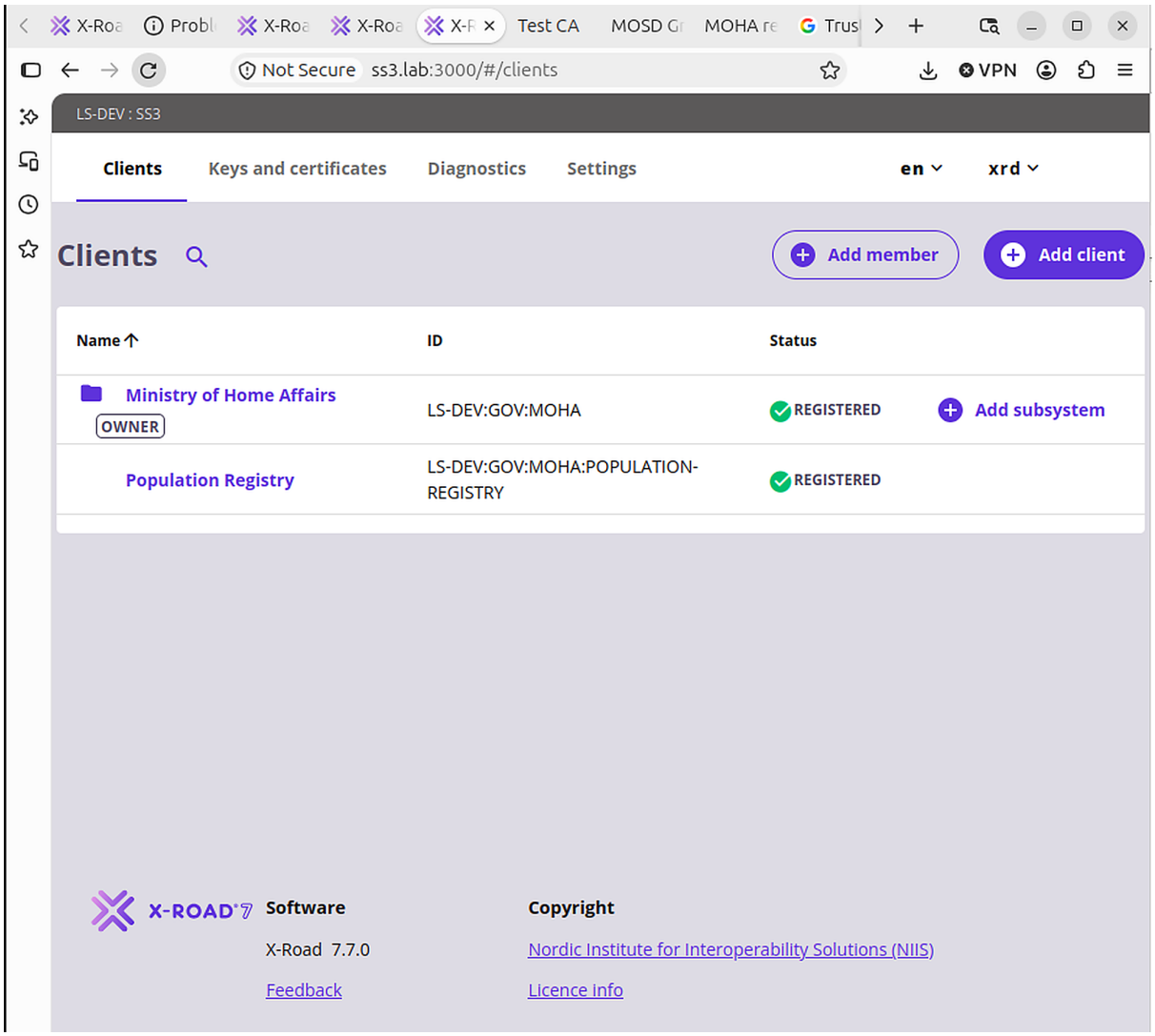


Figure 11. ss3: Population Registry REGISTERED.



Phase 5 — MOHA services

Figure 12. ss3: adding the REST service `population-api` from the OpenAPI 3 description.

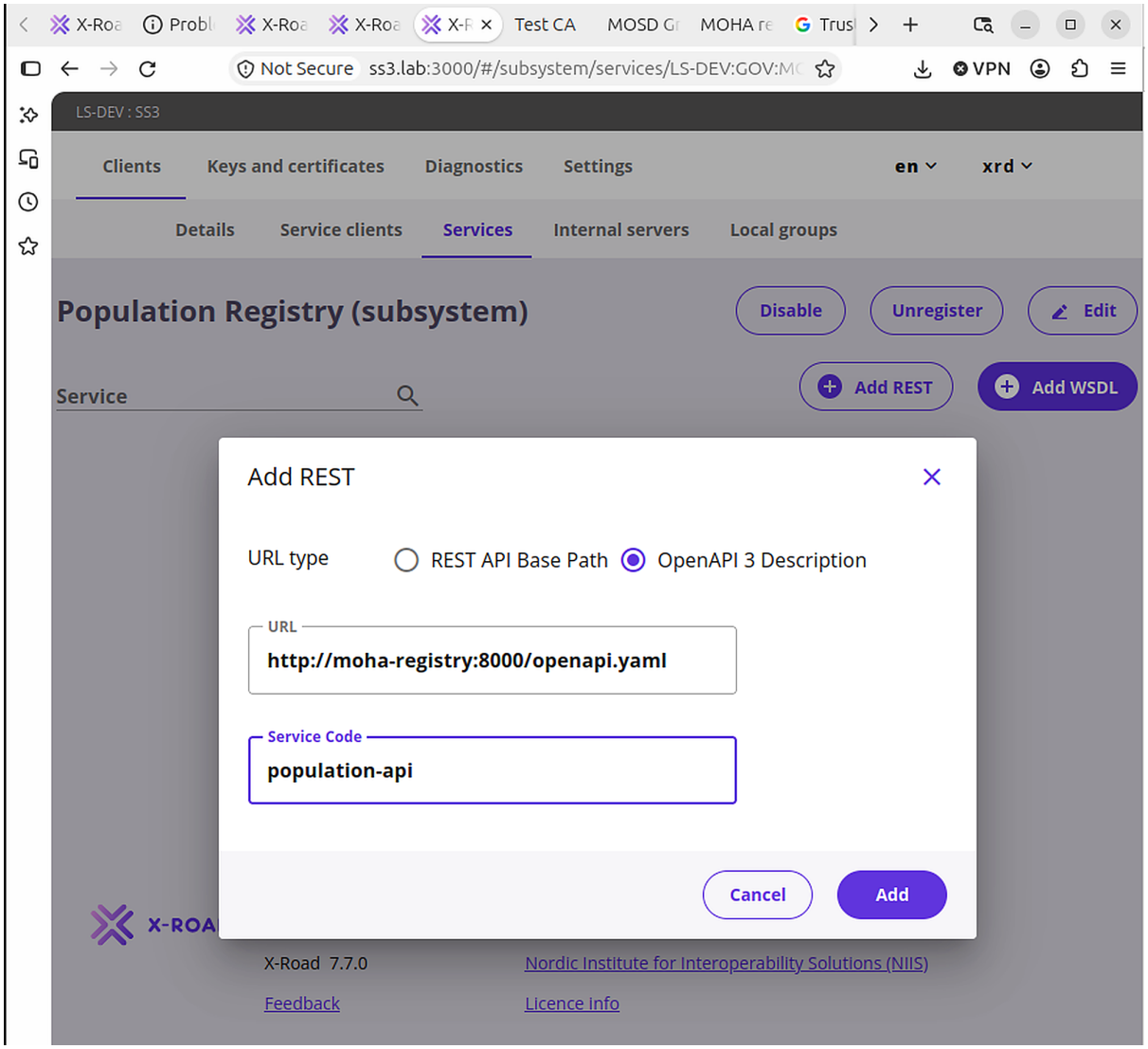
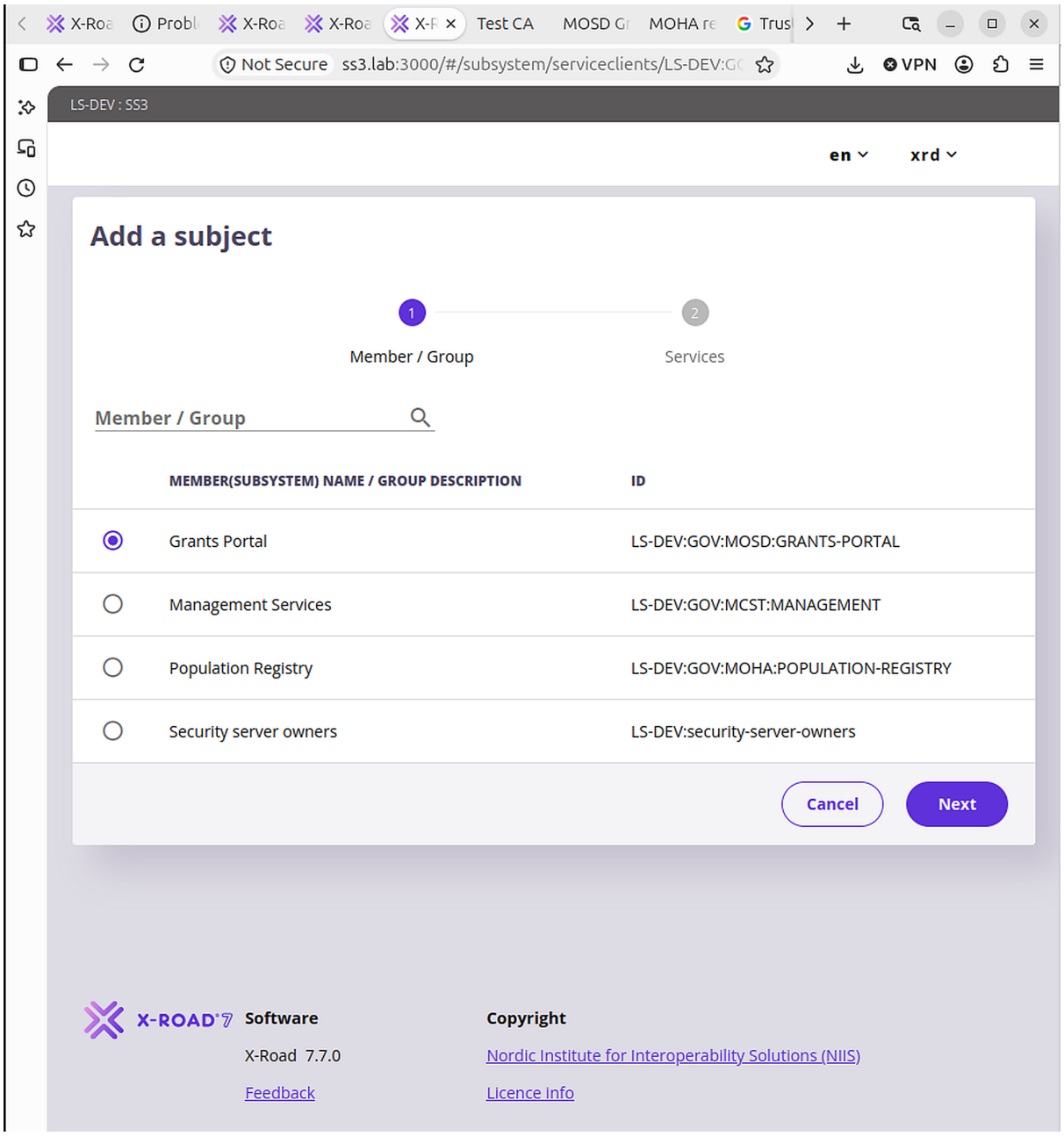


Figure 13. ss3: granting access to `LS-DEV:GOV:MOSD:GRANTS-PORTAL`.



Phase 6 — End-to-end test

Figure 14. Grants Portal: REST getCitizen, HTTP 200 in 594 ms.

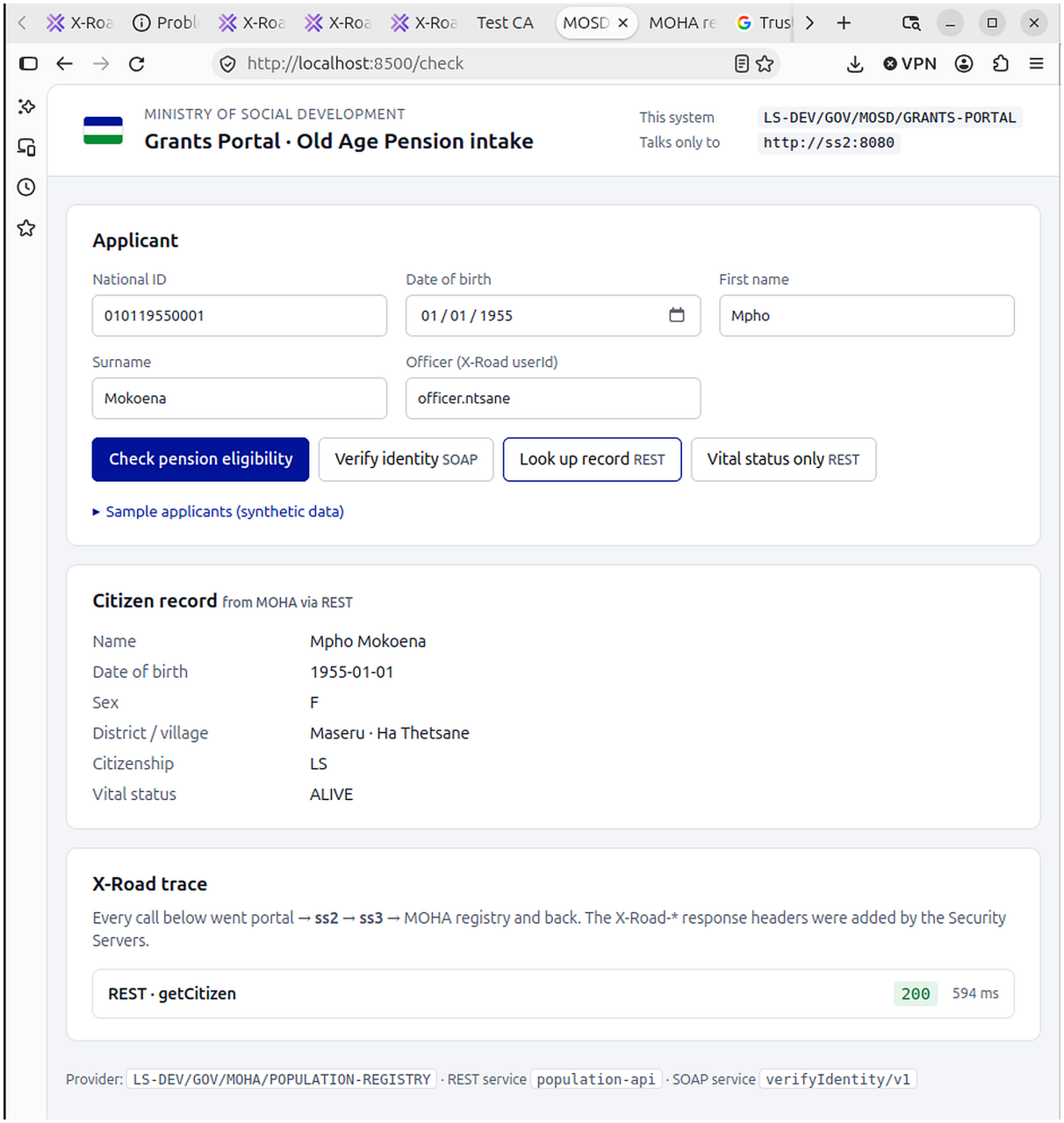


Figure 15. Grants Portal: REST getVitalStatus, HTTP 200 in 187 ms.

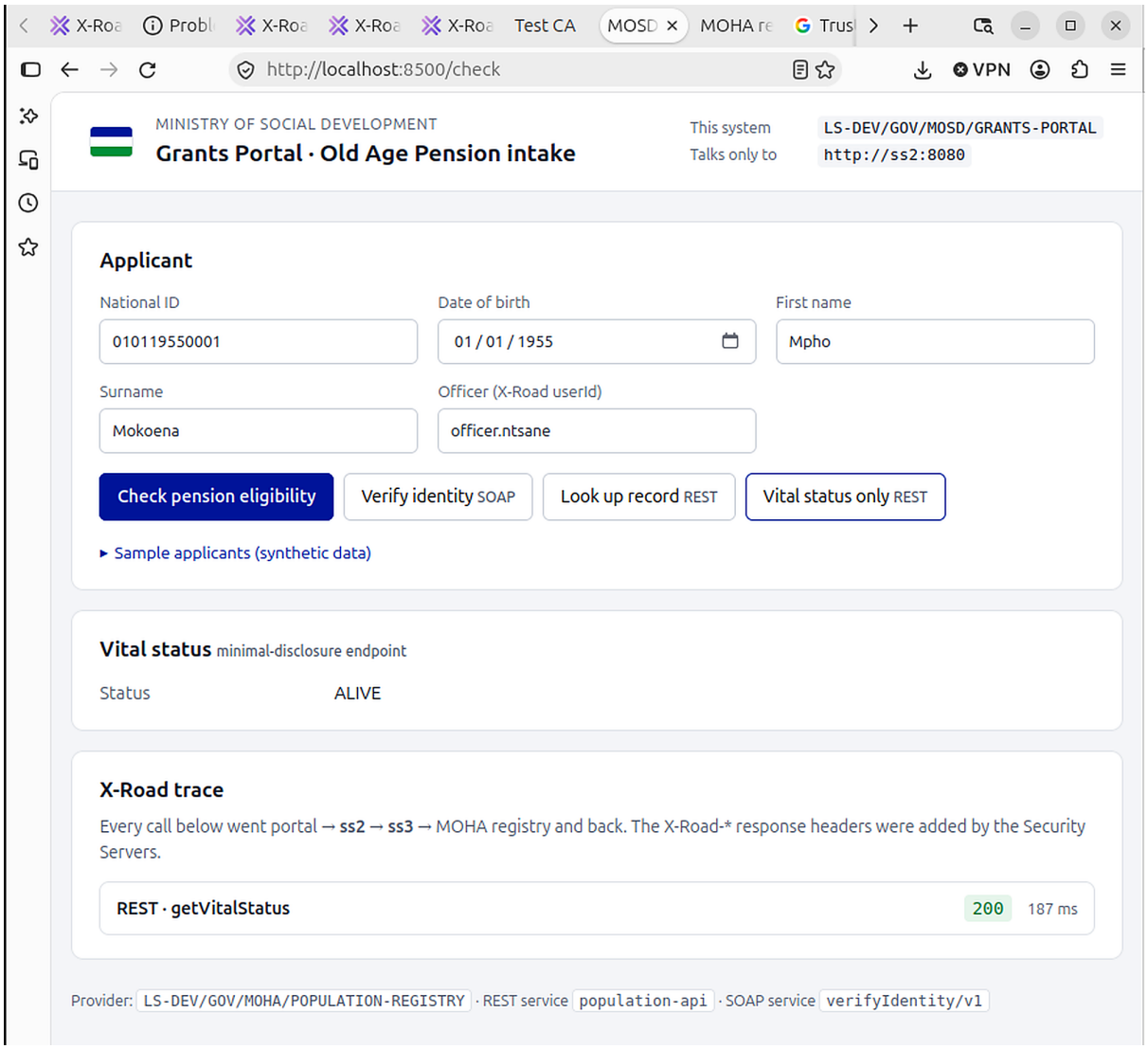


Figure 16. Grants Portal: SOAP verifyIdentity v1, HTTP 200 in 159 ms.

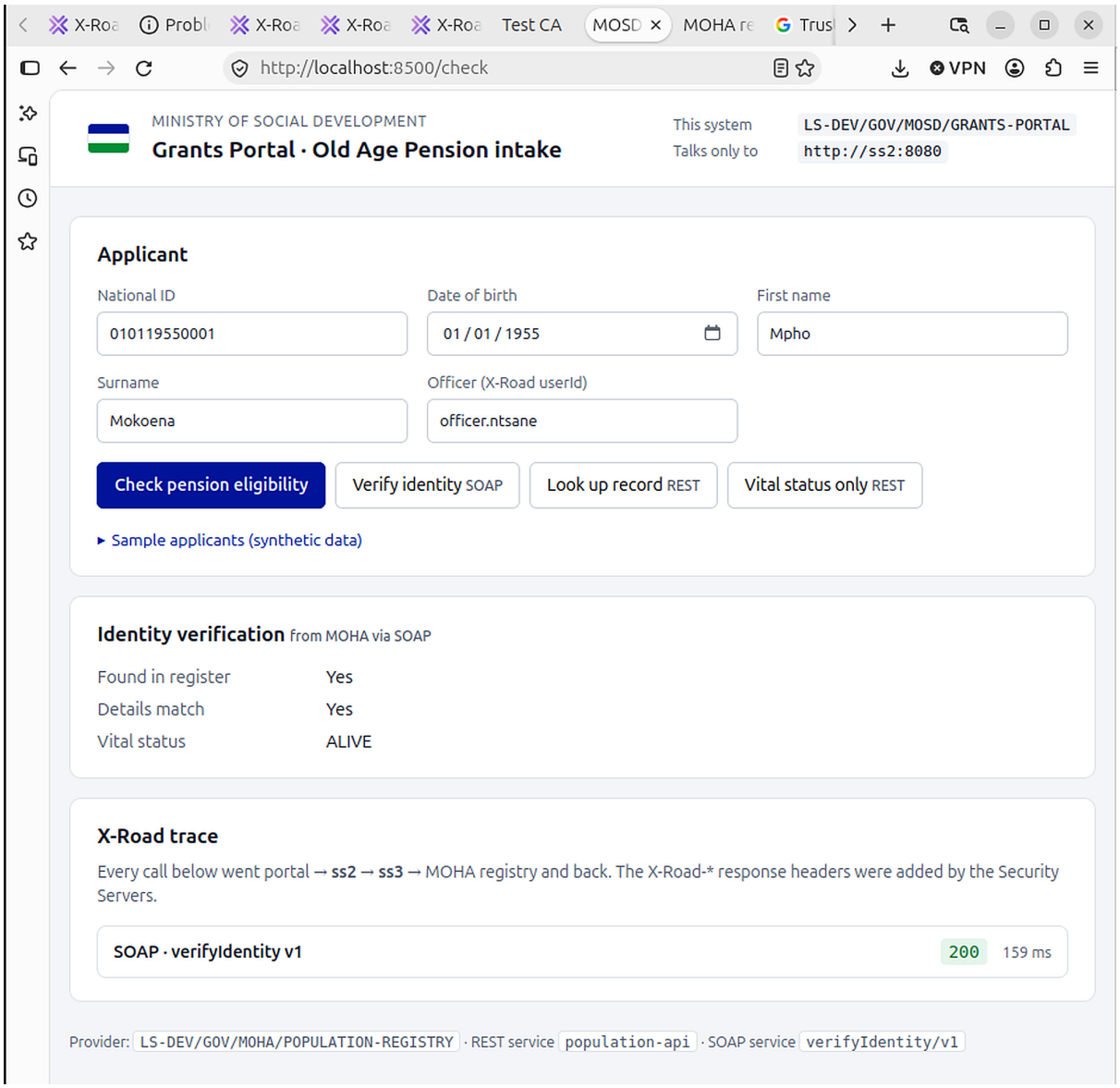
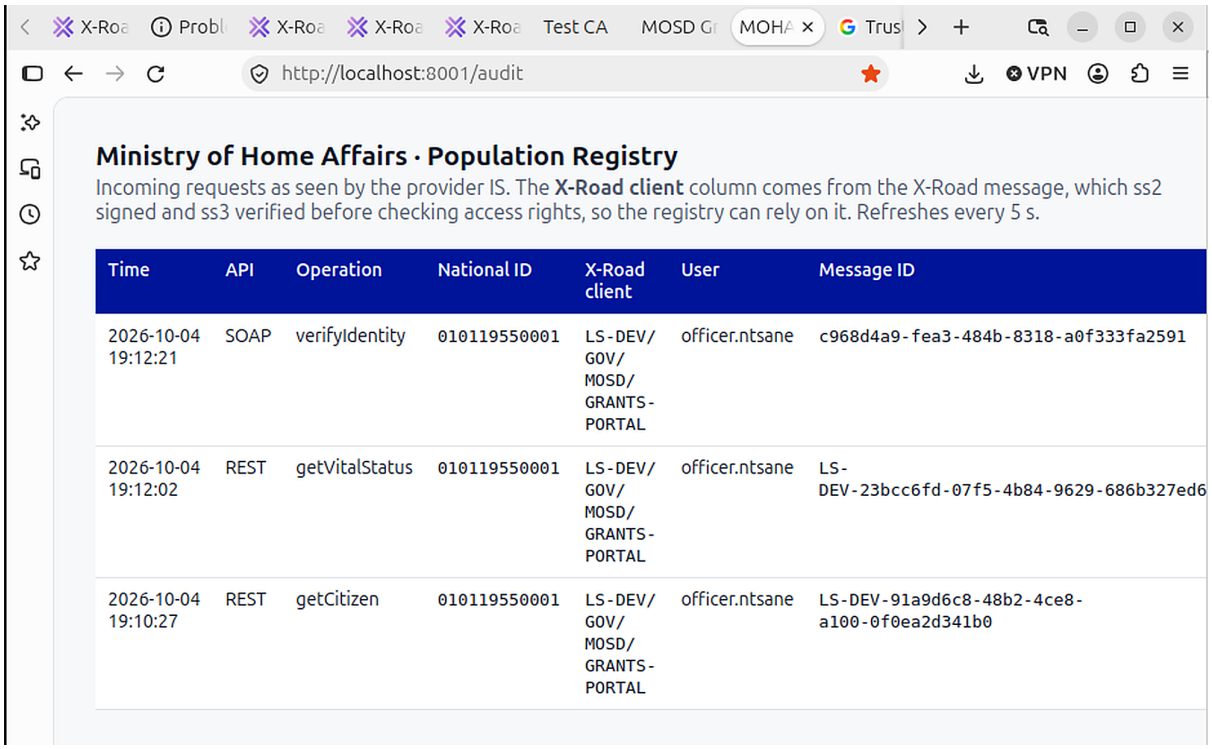


Figure 17. MOHA audit log: all three calls received from GRANTS-PORTAL.



Appendix A. docker-compose.yml

The complete file used for the lab, unchanged. Start everything with `docker compose up -d --build`; stop with `docker compose stop` (never `down -v`, which deletes all 12 volumes).

```
# X-Road local lab: Lesotho government data exchange (LS-DEV)
#
# Based on the NIIS Knowledge Base guide "How to Set Up a Local Test
# Environment Using Docker Compose" (X-Road 7.7.0), extended with two
# information systems:
#
#   mosd-portal   (consumer IS)  Ministry of Social Development Grants Portal
#                 LS-DEV/GOV/MOSD/GRANTS-PORTAL        -> uses ss2
#   moha-registry (provider IS)  Ministry of Home Affairs Population Registry
#                 LS-DEV/GOV/MOHA/POPULATION-REGISTRY  -> behind ss3
#
# Networks model a real deployment:
#   xroad-network  the shared "X-Road network": cs, ca and all Security Servers
#   mosd-lan       MOSD's internal network: only ss2 and the portal
#   moha-lan       MOHA's internal network: only ss3 and the registry
# The portal cannot reach the registry directly; it must go through ss2 -> ss3.
#
# Host ports (bound to 127.0.0.1: reachable only from this laptop.
# Docker publishes ports past ufw, so binding to loopback is what keeps
# the admin UIs and their default xrd/secret login off your Wi-Fi.)
#   4000            Central Server admin UI         https://cs.lab:4000
#   8888            Test CA (signing UI + OCSP)     http://localhost:8888/testca/
#   1000/2000/3000  ss1/ss2/ss3 admin UI            https://ss1.lab:1000 etc.
#                   (cs.lab / ssN.lab are /etc/hosts aliases for 127.0.0.1; see README)
#   1080/2080/3080  ss1/ss2/ss3 consumer HTTP (1443/2443/3443 for HTTPS)
#   8500            MOSD Grants Portal              http://localhost:8500
#   8001            MOHA registry audit/docs        http://localhost:8001/audit
#
# Admin login for cs and all Security Servers: xrd / secret
# Token PIN everywhere: 123456xrd!

services:
  cs:
    image: niis/xroad-central-server:noble-7.7.0
    container_name: cs
    environment:
      XROAD_TOKEN_PIN: 123456xrd!
    ports:
      - 127.0.0.1:4000:4000
    volumes:
      - cs-db-volume:/var/lib/postgresql/16/main
      - cs-conf-volume:/etc/xroad
    networks:
      - xroad-network

  ca:
    image: ghcr.io/nordic-institute/xrddev-testca:latest
    container_name: ca
    ports:
      - 127.0.0.1:8888:8888
    volumes:
      - ca-volume:/home/ca
    networks:
```

```
    - xroad-network

# Management Security Server, owned by the X-Road operator (MCST)
ss1:
  image: niis/xroad-security-server-sidecar:7.7.0
  container_name: ss1
  ports:
    - 127.0.0.1:1000:4000
    - 127.0.0.1:1080:8080
    - 127.0.0.1:1443:8443
  environment:
    XROAD_TOKEN_PIN: 123456xrd!
    XROAD_ADMIN_USER: xrd
    XROAD_ADMIN_PASSWORD: secret
    XROAD_LOG_LEVEL: INFO
  volumes:
    - ss1-db-volume:/var/lib/postgresql/16/main
    - ss1-conf-volume:/etc/xroad
    - ss1-archive-volume:/var/lib/xroad
  networks:
    - xroad-network
  depends_on:
    - cs
    - ca

# Consumer Security Server, owned by MOSD
ss2:
  image: niis/xroad-security-server-sidecar:7.7.0
  container_name: ss2
  ports:
    - 127.0.0.1:2000:4000
    - 127.0.0.1:2080:8080
    - 127.0.0.1:2443:8443
  environment:
    XROAD_TOKEN_PIN: 123456xrd!
    XROAD_ADMIN_USER: xrd
    XROAD_ADMIN_PASSWORD: secret
    XROAD_LOG_LEVEL: INFO
  volumes:
    - ss2-db-volume:/var/lib/postgresql/16/main
    - ss2-conf-volume:/etc/xroad
    - ss2-archive-volume:/var/lib/xroad
  networks:
    - xroad-network
    - mosd-lan
  depends_on:
    - cs
    - ca

# Provider Security Server, owned by MOHA
ss3:
  image: niis/xroad-security-server-sidecar:7.7.0
  container_name: ss3
  ports:
    - 127.0.0.1:3000:4000
    - 127.0.0.1:3080:8080
    - 127.0.0.1:3443:8443
  environment:
```

```
      XROAD_TOKEN_PIN: 123456xrd!
      XROAD_ADMIN_USER: xrd
      XROAD_ADMIN_PASSWORD: secret
      XROAD_LOG_LEVEL: INFO
    volumes:
      - ss3-db-volume:/var/lib/postgresql/16/main
      - ss3-conf-volume:/etc/xroad
      - ss3-archive-volume:/var/lib/xroad
    networks:
      - xroad-network
      - moha-lan
    depends_on:
      - cs
      - ca

  # ---------------------------------------------------------------------------
  # Information systems
  # ---------------------------------------------------------------------------

  # Provider IS: REST (OpenAPI 3) + SOAP (WSDL) population registry
  moha-registry:
    build: ./moha-registry
    image: ls-lab/moha-registry:1.0
    container_name: moha-registry
    ports:
      # Only so you can open /audit and /docs from your browser.
      # In production a provider IS is never exposed like this.
      - 127.0.0.1:8001:8000
    networks:
      - moha-lan
    restart: unless-stopped

  # Consumer IS: grants officer web app
  mosd-portal:
    build: ./mosd-portal
    image: ls-lab/mosd-portal:1.0
    container_name: mosd-portal
    environment:
      XROAD_SS_URL: http://ss2:8080
      XROAD_CLIENT: LS-DEV/GOV/MOSD/GRANTS-PORTAL
      XROAD_PROVIDER: LS-DEV/GOV/MOHA/POPULATION-REGISTRY
      XROAD_REST_SERVICE: population-api
      XROAD_SOAP_SERVICE: verifyIdentity
      XROAD_SOAP_VERSION: v1
    ports:
      - 127.0.0.1:8500:8000
    networks:
      - mosd-lan
    depends_on:
      - ss2
    restart: unless-stopped

volumes:
  cs-db-volume:
    name: cs-db-data
  cs-conf-volume:
    name: cs-conf-data
  ca-volume:
```

```
    name: ca-data
  ss1-db-volume:
    name: ss1-db-data
  ss1-conf-volume:
    name: ss1-conf-data
  ss1-archive-volume:
    name: ss1-archive-data
  ss2-db-volume:
    name: ss2-db-data
  ss2-conf-volume:
    name: ss2-conf-data
  ss2-archive-volume:
    name: ss2-archive-data
  ss3-db-volume:
    name: ss3-db-data
  ss3-conf-volume:
    name: ss3-conf-data
  ss3-archive-volume:
    name: ss3-archive-data

networks:
  xroad-network:
    name: xroad-network
    driver: bridge
  mosd-lan:
    name: mosd-lan
    driver: bridge
  moha-lan:
    name: moha-lan
    driver: bridge
```